



COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM(S) TO BE EVALUATED BS-CS Spring 2026

Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	AL-2002
Course Title	Artificial Intelligence
Credit Hours	1
Course Instructor	
Prerequisites by Course(s) and Topics	Data Structures (CS-2001) Object Oriented Programming (CS-1004)
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Lab, project, and assignments will cover the programming part of the course. Breakdown of Course Work (Total): Midterm 1 15% Lab Work 15 30% Project 1 20% Final 1 35%
Grading Policy	Absolute grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterms/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.
Course Plagiarism Policy	Plagiarism in project or midterms/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Lab Instructors (if any)	Ms. Zille Huma, Mr. Imran Khan, Mr. Ali Umer
Lab Coordinator	Ms. Zille Huma
URL (if any)	
Current Catalog Description	This course introduces students to the basic knowledge representation, problem solving, and learning methods of artificial intelligence. Upon completion, students should be able to develop intelligent systems by assembling solutions to concrete computational problems; understand the role of knowledge representation, problem solving, and learning in intelligent-system engineering; and appreciate the role of problem solving, vision, and

	language in understanding human intelligence from a computational perspective.
Textbook (or Laboratory Manual for Laboratory Courses)	• Artificial Intelligence: A Modern Approach, 3rd ed. S. Russell and P. Norvig, Prentice Hall, 2010. • Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th ed. G. Luger, Addison Wesley, 2009 • AI Algorithms, Data Structures, and Idioms in Prolog, Lisp and Java, G. Luger and W. Stubblefield, Addison Wesley, 2009 • Artificial Intelligence: A Systems Approach. M. Tim Jones, Infinity Science Press, 2008
Reference Material	Supporting website: http://aima.cs.berkeley.edu

Course Learning Outcomes	A. Course Learning Outcomes (CLOs)			
	On successful completion of this course students will have to know how of:			
	1. Explain how AI enables intelligent behavior in real-world applications such as game playing, autonomous vehicles, and robotic systems.			
	2. Apply classical AI techniques, including uninformed and informed search algorithms, minimax, and neural networks, to solve structured problems.			
	3. Design basic AI-based solutions for real-world problems using appropriate techniques such as reasoning, tracking, or robot localization.			
	B. Program Learning Outcomes			
	For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.			
	1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals	
	2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements	✓
	3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines	✓
4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations	✓	
5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations		
6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings		
7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions		
8	Computing Professionalism	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and		

		and Society	the consequential responsibilities relevant to professional computing practice								
	9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice								
	10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional								

C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)											
		PLOs									
		1	2	3	4	5	6	7	8	9	10
CLOs	1		✓								
	2			✓							
	3				✓						

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Topics to be covered:			
	List of Topics	No. of Weeks	Contact Hours	CLO(s)
	Introduction to AI	1	3	1
	Intelligent Agents: Problem Solving, Agent types, Agent environments	1	3	1,3
	Search problem formulation, Search space definition, Tree Search, Graph Search, Uninformed searches - BFS, DFS, UCS	1	3	2
	Informed Search: Heuristic Function, Greedy Best First Search, and A* search. Problems with Search	1	3	2,3
	Local Search: Hill Climbing, Beam search, simulated annealing	1	3	2
	Optimization: Genetic Algorithm.	1	3	2
	Constraint Satisfaction Problems Back-tracking search for CSPs, Problem structure and problem decomposition, Local search for CSPs, Cryptarithmic Problem. Forward checking, AC3, MRV, LCV	2	6	2,3
	Game as Search Problems, Adversarial search Minimax and alpha beta pruning.	1	3	1,2
	Intro to Machine Learning, Supervised Learning, KNN, Decision Tree	1	3	2
	Unsupervised Learning, K-Means, Hierarchical Clustering	1	3	2
	Total	15	45	
Laboratory Projects/Experiments Done in the Course				
Programming Assignments Done in the Course	Yes, All the assignments were programming based, the language should be python.			

Class Time Spent on (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	50	25	20	5
Oral and Written Communications	Every student is required to submit at least <u> 4 </u> written reports of typically <u> 4-5 </u> pages and to make <u> 1 </u> oral presentation of typically <u> 20 </u> minute's duration.			

Lab/ Practical Component of the course

Weeks	Contents/Topics	Assessment Items (Case Study/ Exercise Assignment/ Quiz etc.)	CLO Mapping
Week-01	Introduction to Python Language	Lab Task	1
Week-02	OOP concepts, Lambda Functions, NumPy	Lab Task	1,2
Week-03	Intelligent Agents	Lab Task	1,2
Week-04	Uninformed Searches (BFS, DFS, UCS) & Greedy Search	Lab Task	2
Week-05	Informed Searches (A*, Hill climbing)	Lab Task	2
Week-06	Local Search Algorithms (Beam Search, Simulated annealing)	Lab Task	2
Week-07	Genetic Algorithm	Lab Task	2,3
Week-08	Constraint Satisfaction Problems	Lab Task	2
Week-09	Minimax AI	Lab Task	1,2
Week-10	Alpha Beta Pruning	Lab Task	2
Week-11	Data Preprocessing	Lab Task	2



Week -12	Classification (KNN)	Lab Task	2
Week 13	Decision Tree	Lab Task	2
Week-14	K-Means Clustering	Lab Task	2
Week-15	Neural Networks	Lab Task	1,2